Hall Ticket Number:

MR-22

Code: B425CS2S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025 / January 2026

Discrete Mathematics

(Common to CSE, IT and CSM)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

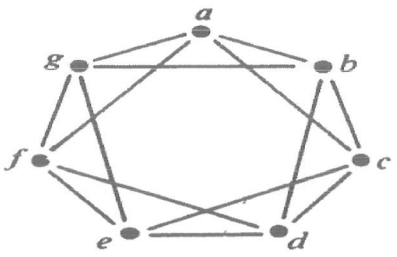
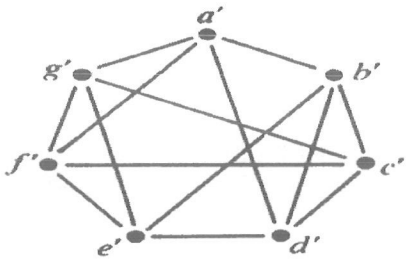
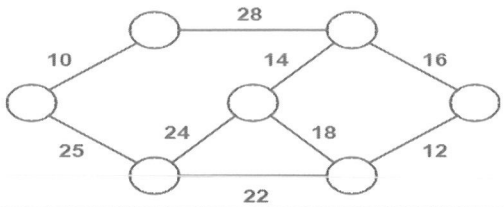
Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What are well formed formulas?	1	1	1	1
1. b)	Define CNF.	1	2	1	1
1. c)	What is the difference between a subset and a proper subset?	1	2	2	2
1. d)	Give an example of a Bijective function.	1	1	2	2
1. e)	Define Algebraic system.	1	1	3	1
1. f)	How do we denote Boolean algebra?	1	2	3	1
1. g)	Define permutation.	1	1	4	1
1. h)	How many 10 digit numbers are there with exactly six 1's?	1	2	4	2
1. i)	Define order and size of a Graph.	1	2	5	1
1. j)	What is the chromatic number of a graph?	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Show that **((P ∨ Q) ∧ ¬ (¬P ∧ (¬ Q ∨ ¬R))) ∨ (¬P ∧ ¬Q) ∨ (¬P ∧¬ R)** is a tautology.	5	2	1	3
2. b)	Prove that **∃x M(x)** follows logically from the premises **(x) (H(x) → M(x))** and **∃x H(x).**	5	1	1	3
	OR				
2. c)	Obtain PDNF for **(¬P → R) ∧ (Q⇄P).**	5	1	1	2
2. d)	Define Predicate. Discuss about Free and Bound variables with examples.	5	1	2	1
3. a)	Let A= {1,2,3,4}, B= {3,4,5,6}, C= {1,3,5} Find: a) A∪B, b) A∩B, c)(A∪B)∩C, d)(A∪B∪C), e) (A∩B∩C).	5	2	2	2
3. b)	Define Binary operation. Discuss some general properties of Binary operations.	5	2	2	1
	OR				
3. c)	Draw Hasse diagram for $[I_{12}, \|]$.	5	3	2	2
3. d)	State and explain about any four frequently used Primitive Recursive Functions.	5	4	2	1
4. a)	Discuss about Homomorphism of Semigroups and Monoids.	5	2	3	1
4. b)	Prove that in any lattice (L,∨,∧), the following identity holds: a∨(a∧b)=a and a∧(a∨b)=a.	5	2	3	2
	OR				
4. c)	Give an example of an algebraic system that is associative but not commutative. Prove both properties using examples.	5	2	3	2
4. d)	Describe the concepts Subalgebra and Direct Product.	5	1	3	1

P.T.O.

5. a)	Consider the word TRIANNUAL. (i) How many arrangements are there of these 9letters? (ii) How many 9-letter words are there with the letters T, I, and U Separated by exactly 2 of the other letters? (iii) How many 6-letter words can be formed from the letters of TRIANNUAL with no N's?	5	3	4	3
5. b)	What is the coefficient of x^3y^7 in $(x + y)^{10}$ and in $(2x - 9y)^{10}$?	5	2	4	2
	OR				
5. c)	Evaluate the sum (i) $1 + 2 + 3 + ... + n$. (ii) $1^2 + 2^2 + 3^2 + ... + n^2$.	5	2	4	2
5. d)	In a survey of 800 voters, the following information was found: 300 were college educated, 260 were from high-income families, 325 were registered Democrats, 184 were college educated and from high-income families, 155 were college educated and registered Democrats, 165 were from high-income families and were registered Democrats, 94 were college educated, from high-income families, and were registered Democrats. Let E = {voters who were college educated} I= {voters who were from high-income families} D = {voters who were registered Democrats}. Draw a Venn diagram and list the number of elements in the 8 different regions of the diagram.	5	2	4	3
6. a)	Identify whether the following graphs G and G' are isomorphic or not. Justify.  Graph G  Graph G'	5	3	5	2
6. b)	Differentiate between Euler graphs and Hamiltonian Graphs.	5	3	5	1
	OR				
6. c)	What is a Planar Graph? Find the number of regions in Graph G, given vertices are 20 and degree of each vertex is 3.	5	1	5	2
6. d)	Construct the minimum spanning tree (MST) for the given graph using Kruskal's Algorithm.  	5	3	5	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome